

SmolVLM

Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

256M

Smallest Model

<1GB

GPU RAM

300×

vs Ldefics-80B

Large VLMs Are Powerful but Impractical for Edge Deployment



Massive Memory Footprint

Molmo-A1B-7B requires 27.7 GB GPU RAM — impossible on mobile devices



Compute-Heavy Inference

State-of-the-art VLMs need high-end GPUs, blocking edge deployment



Long Token Sequences

High-res images generate thousands of visual tokens, overwhelming small LMs

GPU RAM Comparison

SmoIVLM-256M

0.8 GB

SmoIVLM-500M

1.2 GB

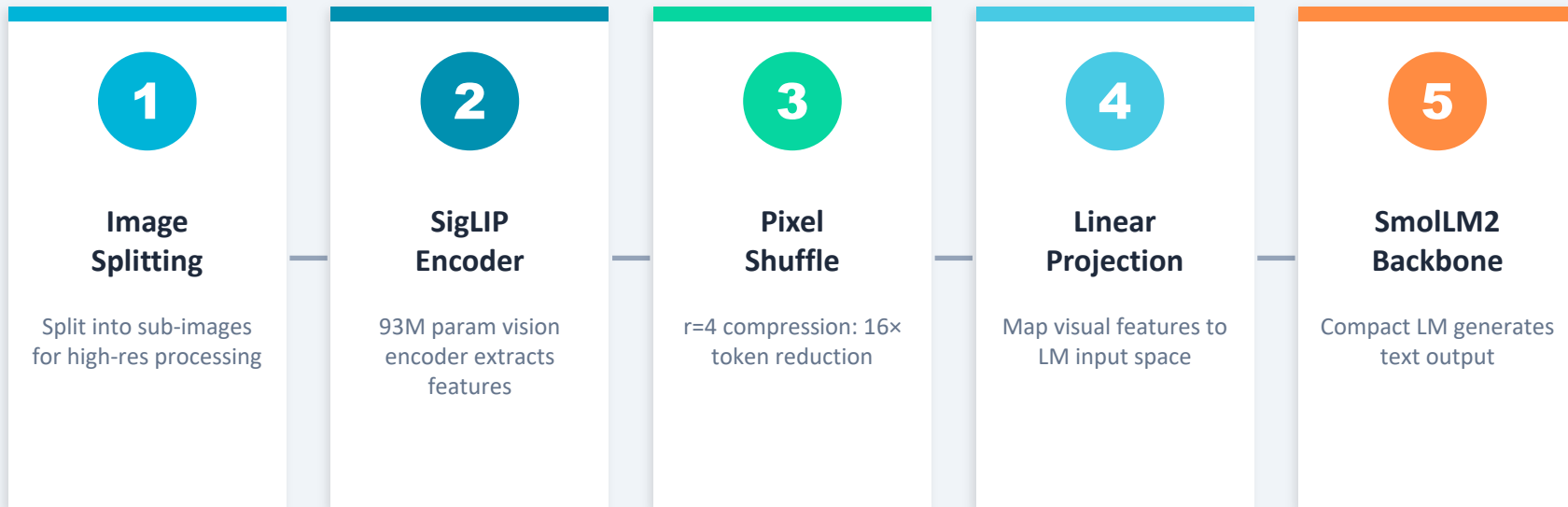
SmoIVLM-2.2B

4.9 GB

Molmo-A1B-7B

27.7 GB

SmolVLM Architecture Pipeline

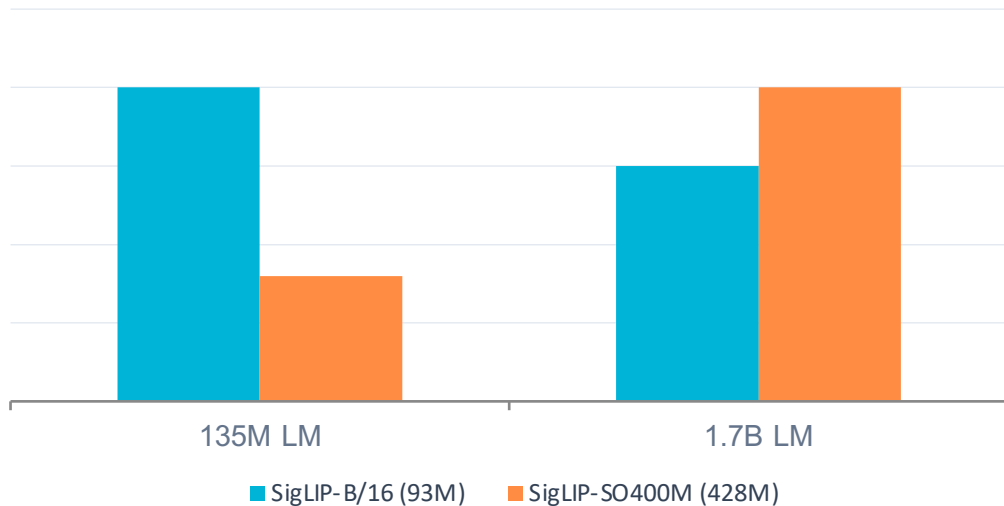


Video: frames sampled and processed through the same pipeline

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1: Balanced encoder-LM compute allocation is critical for small models

Relative Performance by Encoder × LM Size



93M SigLIP-B/16 outperforms the 4.6× larger SigLIP-SO400M (428M) when paired with 135M LM



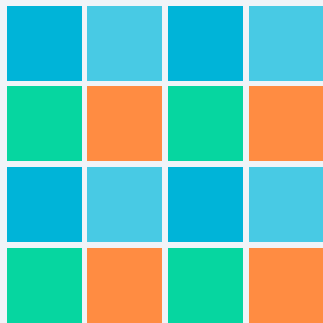
Larger encoder only justifies its 10% parameter overhead at 1.7B LM scale



Takeaway: match encoder capacity to LM scale — oversized encoders waste parameters in compact VLMs

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16×

Input Feature Map

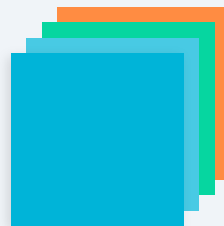


$H \times W$ spatial tokens



Pixel
Shuffle
 $r=4$

Output: Compressed Tokens



$H/r \times W/r$ tokens
 $r^2 \times C$ channels

Token Reduction

$r=2$

4×

Large VLMs

$r=4$

16×

Small VLMs

Pixel shuffle rearranges spatial features into channels, reducing visual tokens by r^2 . Small VLMs benefit from $r=4$, easing attention overhead and improving long-context modeling.

Extending Context from 2K to 16K Tokens Unlocks Higher Resolution

Finding 2: Performance improves significantly with extended context length

2K tokens — RoPE
base: 10K

8K tokens — RoPE base: 273K

16K tokens — RoPE base: 273K



RoPE Base Adjustment

Increased from 10K → 273K to support longer positional encoding



Mixed-Length Training

Fine-tuned on combined long/short-context data for robust generalization



Model-Specific Context

2.2B model: 16K tokens
Smaller variants: 8K tokens



Synergy with Pixel Shuffle

16× token reduction + extended context enables high-res image input within budget

Benchmark Performance: Competitive at Sub-1GB to 4.9GB RAM

Category	Benchmark	SmoIVLM 256M	SmoIVLM 500M	SmoIVLM 2.2B	Best Efficient OS Baseline
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7%
	AI2D	46.4%	59.2%	70.0%	71.0%
	ChartQA	55.6%	62.8%	68.7%	48.0%
	TextVQA	50.2%	60.2%	73.0%	61.5%
	DocVQA	58.3%	70.5%	80.0%	77.7%
	ScienceQA	73.8%	80.0%	89.6%	87.5%
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9%
	MathVista	35.9%	40.1%	51.5%	37.6%
	MMStar	34.6%	38.3%	46.0%	43.1%
Video	Video-MME	33.7%	42.2%	52.1%	45.0%
	MLVU	40.6%	47.3%	55.2%	48.2%
Average	All Benchmarks	44.0%	51.0%	59.8%	—

RAM Usage: SmoIVLM-256M: 0.8 GB | SmoIVLM-500M: 1.2 GB | SmoIVLM-2.2B: 4.9 GB | Molmo-A1B-7B: 27.7 GB

Real-Time On-Device Inference on Apple Silicon



HuggingSnap iOS App

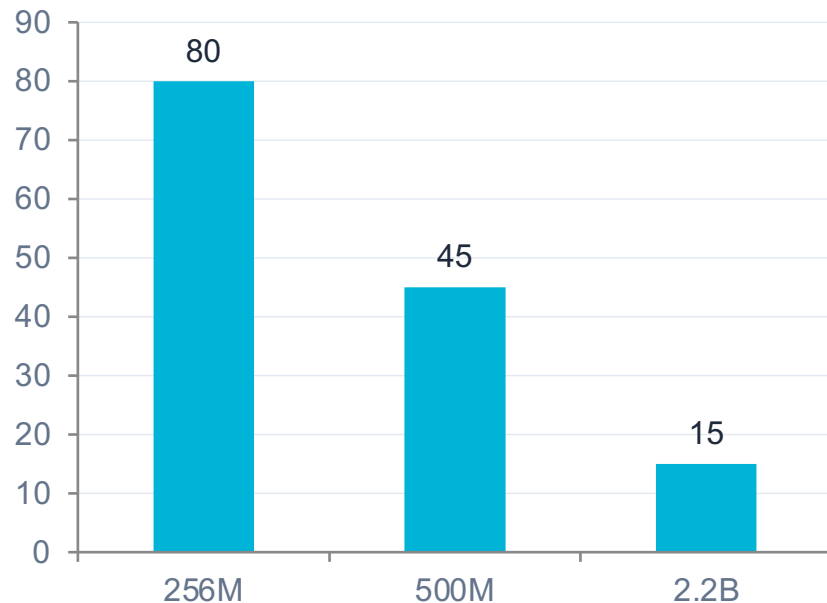
- Fully local inference — no cloud dependency
- Processes photos and videos on-device
- Runs SmolVLM-256M with <1GB RAM
- Open-source: weights, code, and app

MacBook Pro M1 Max 64GB ~80 tok/s

MacBook Air M3 16GB ~55 tok/s

Batch=64, high-end ~1000 tok/s

Peak Throughput on M1 Max (tokens/sec)



Throughput scales roughly linearly with batch size for the smallest model

Key Takeaways: Design Principles for Efficient Multimodal Models



Balance Encoder–LM Compute

Match encoder to LM scale — 93M encoder beats 428M for small LMs



Extend Context Length

Scale 2K→16K via RoPE adjustment for higher image resolution



Aggressive Token Compression

Pixel shuffle $r=4$ gives 16× reduction, critical for attention efficiency



Sub-1GB Inference Is Achievable

SmolVLM-256M: 0.8 GB RAM, outperforming 300× larger LLaVA-80B



Fully Open-Source Release

All weights, datasets, code, and HuggingSnap mobile app released

SmolVLM — Competitive multimodal AI, from the cloud to your pocket.